



Lidl 2020 MEP Playbook

HVAC, Controls and Electrical Install Assistance Guide

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Guide Prepared by HQ CM

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This guide is to assist the installing contractors with Mechanical, Electrical and BMS Controls. The guide is to address the most common issues found during testing and does not cover the complete scope of work. If assistance needed with clarification please reach out to the Construction Manager for assistance.

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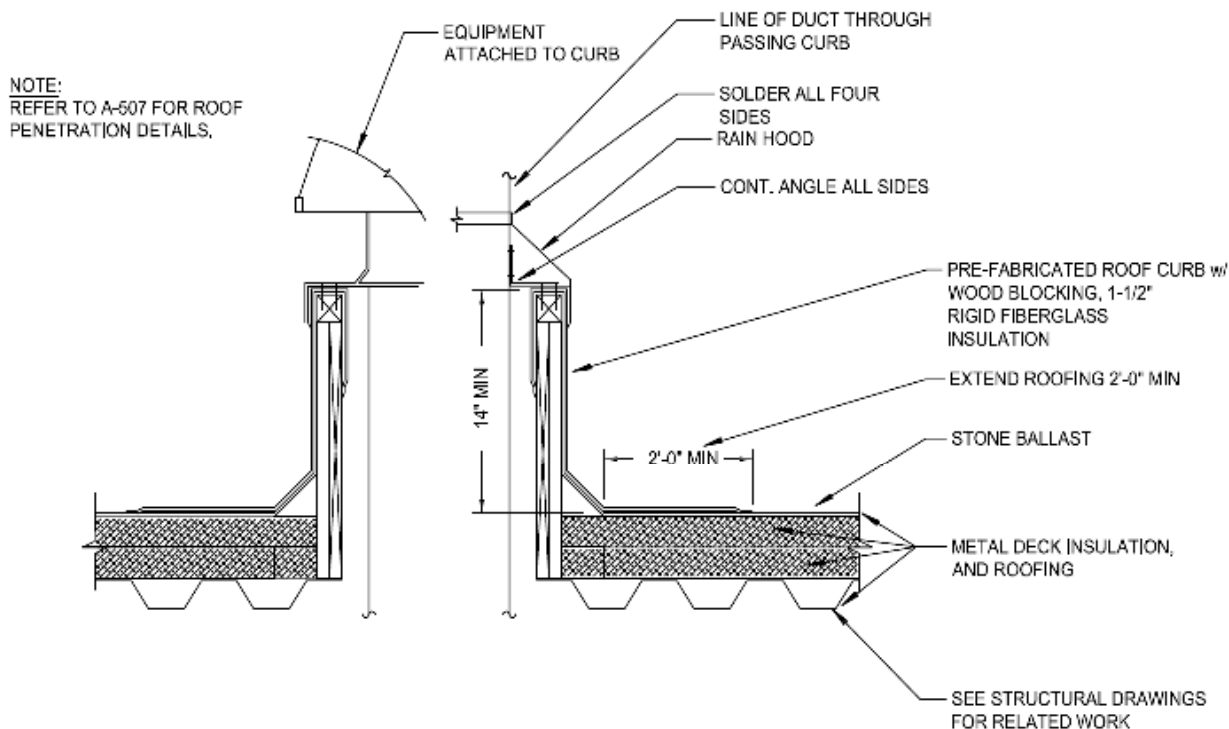




1. (MECHANICAL) HVAC ROOF CURB AND EQUIPMENT INSTALL

a. Lennox, Munters, AC Split Condensing Units and Kitchen Exhaust Fan.

All curbs to be 14" above Deck Insulation, not 14" from roof deck or joist.



6 ROOF CURB DETAIL

NTS

On slanted roof curb to be 14" on high side and 12" on low side. Field Verify.

On flat roof all curbs to be 14" on all sides. Non-Standard site or existing equipment may be exemption. AC Split condensing units plans may call for a pad instead of a curb.

Condensing units are to be installed side by side as examples below show and spaced apart per Manufacturer design. Any other installation type without written Construction Manager approval will be corrected at HVAC Contractor expense.



b. AC Split Units clearances and orientation

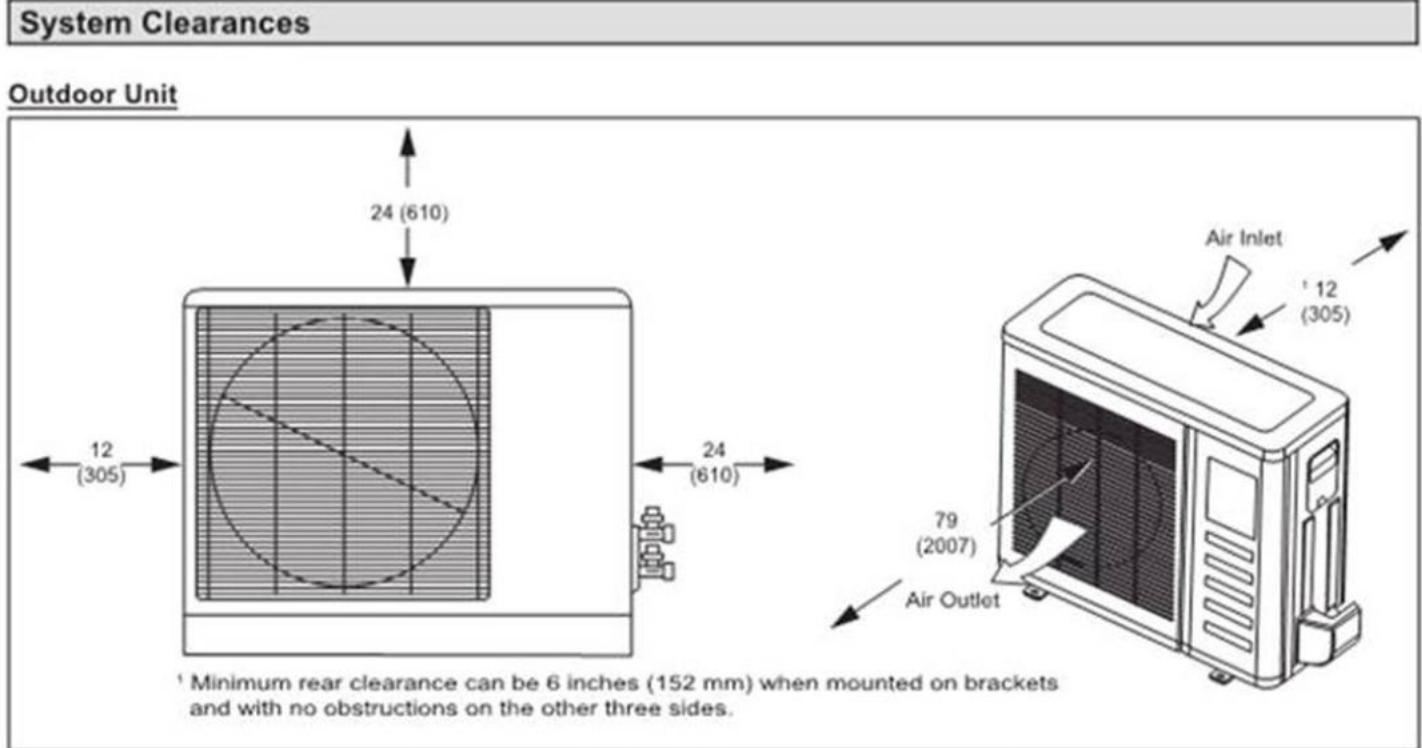
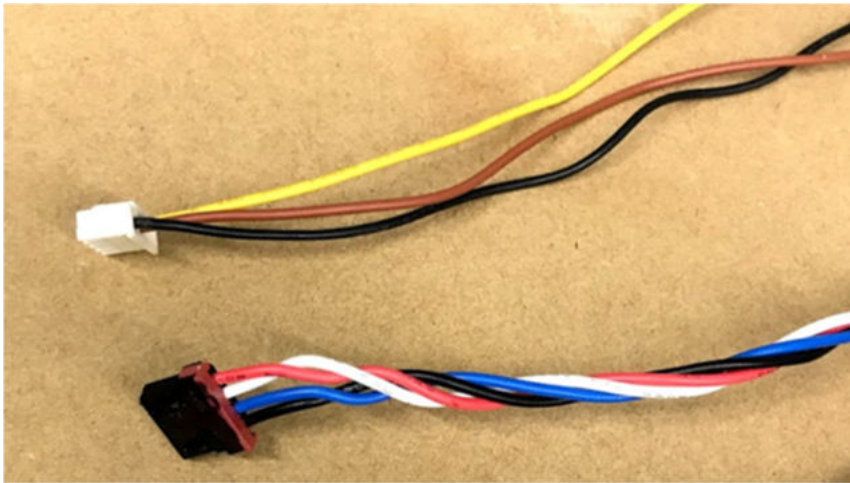


Figure 2. Outdoor Unit Clearances - Inches (mm)





Mini-split wall-mounted indoor units are shipped with a cable for use with connecting to XYE terminal. Do not discard.





c. Munters and Lennox Startup requirements.



ATTENTION CONTRACTORS

START UP CHECKLIST

Please see that the items related to this unit are complete prior to scheduling the factory technician for startup.

1. Piping for gas and required gas regulators shall be completed per U.M.C., state and or local code requirements and as per plans and specifications.
2. Electrical service shall be completed as per N.E.C., state and or local code requirements and as per plans and specifications.
3. Duct connections shall be completed as per U.M.C., state and or local requirements and as per plans and specifications.
4. Ensure that all duct mounted sensors are installed as per plans and specifications.
5. Incoming BMS control wiring shall be completed as per N.E.C., state and or local code requirement and as per plans and specifications.
6. Piping for refrigerant shall be completed as per A.S.H.R.A.E., U.M.C., stated and or local code requirements and as per plans and specifications.
7. Piping for chilled, condenser, and condensate water shall be completed as per A.S.H.R.A. E., U.M.C., state and or local code requirements and as per plans and specifications.
8. Piping for steam shall be completed as per A.S.H.R.A.E., U.M.C., and state and or local code requirements and as per plans and specifications.
9. Install approved "P" trap of proper depth on cooling coil condensate drain. Install approved cap on any other drain connection for cooling coil. Trap shall be completed per U.M.C., state and or local code requirements and as per plans and specifications.
10. Install all devices such as heat reclaim, evaporator, condenser, and remote air handlers as per A.S.H.R.A.E., U.M.C., state and or local requirements, and as per plans, specifications and in accordance with Munters specifications.
11. Install all control devices such as thermostats, humidistats, , pressure sensors, damper actuators, photoelectric controls, remote panels, remote sensors, etc. as per N.E.C., state and or local code requirements, plans, specifications and in accordance with Munters specifications.
12. Install all weather hoods; mist eliminators, covers, louvers, or panels in accordance with Munters specifications. Ensure that all devices are sealed with an approved silicone type of sealant, and that all openings are made water tight.
13. Ensure that the react outlet cover opens and closes freely.

Please understand that any additional expenses resulting from the equipment not being installed will be the responsibility of your Company at a rate of \$108.00 per hour plus related expenses.

Please call Munters Corporation with related questions and notify Munters service coordinators two to four weeks in advance of startup at 1-800-229-8557.





Lennox Mechanical Contractor Equipment Startup

In regard to the startup of Lennox HVAC product, please see below excerpt from page 5 of our installations manual. This breaks down what is needed to keep from voiding the Lennox Limited Product Warranty if used during the construction phase.

Use of the unit as a construction heater or air conditioner is not recommended during any phase of construction. Very low return air temperatures, harmful vapors and operation of the unit with clogged or misplaced filters will damage the unit.

If this unit has been used for heating or cooling of buildings or structures under construction, the following conditions must be met or the warranty will be void:

- The vent hood must be installed per these installation instructions.
- A room thermostat must control the unit. The use of fixed jumpers that will provide continuous heating or cooling is not allowed. (Unit may run in Prodigy Stand Alone mode)
- A pre-filter must be installed at the entry to the return air duct.
- The return air duct must be provided and sealed to the unit.
- Return air temperature range between 55°F (13°C) and 80°F (27°C) must be maintained.
- Air filters must be replaced and pre-filters must be removed upon construction completion.
- The input rate and temperature rise must be set per the unit rating plate.
- The heat exchanger, components, duct system, air filters and evaporator coil must be thoroughly cleaned following final construction clean-up.

The unit operating conditions (including airflow, cooling operation, ignition, input rate, temperature rise and venting) must be verified according to these installation instructions.





Q4. How can I setup the unit to operate temporarily using the factory-installed return air sensor, prior to installation of the A2 zone sensor? (How do I set the unit up for temporary heating and cooling?)

You can also refer to the HowTo video at the following link for help:

<http://www.lennoxcommercial.com/landing/prodigy/how-to-videos.asp>

In order to temporarily operate the unit with the installed return air sensors, go to

SETUP > NETWORK INTERGRATION > RTU STANDALONE press the **SAVE** button. Room sensor back up mode will appear and use the adjust value buttons until you see **= RETURN AIR BACKUP** and press the **SAVE** button. Continue through and set the desired temperature set points, until the display returns to the **SETUP** menu.

Since the unit is now running off of the factory installed return air sensor we need to make the blower run continuous so it has a good sample of the air temperature. To do this go to;

SETTINGS > RTU OPTION> EDIT PARAMETER=154

Press and hold the adjust up button until a parameter **154** appears and press **SAVE**. Press the adjust up button until **=ON-CONTINUOUS 1** is displayed and press **SAVE**. Press the back button as many times as needed to return to the unit status screen. After a five minute time delay the unit will start operation based on the temperature read at the return air sensor.

(Edit parameter not required for blower operations change. It can be done through settings/blower menu. Also, blower should not have to run continuously when run in standalone)

CAUTION: The use of this unit as a construction heater or air conditioner is not recommended during any phase of construction. Very low return air temperatures, harmful vapors and operation of the unit with clogged or misplaced filters will damage the unit. If this unit has been used for heating or cooling of buildings or structures under construction, the following conditions must be met or the warranty will be void:

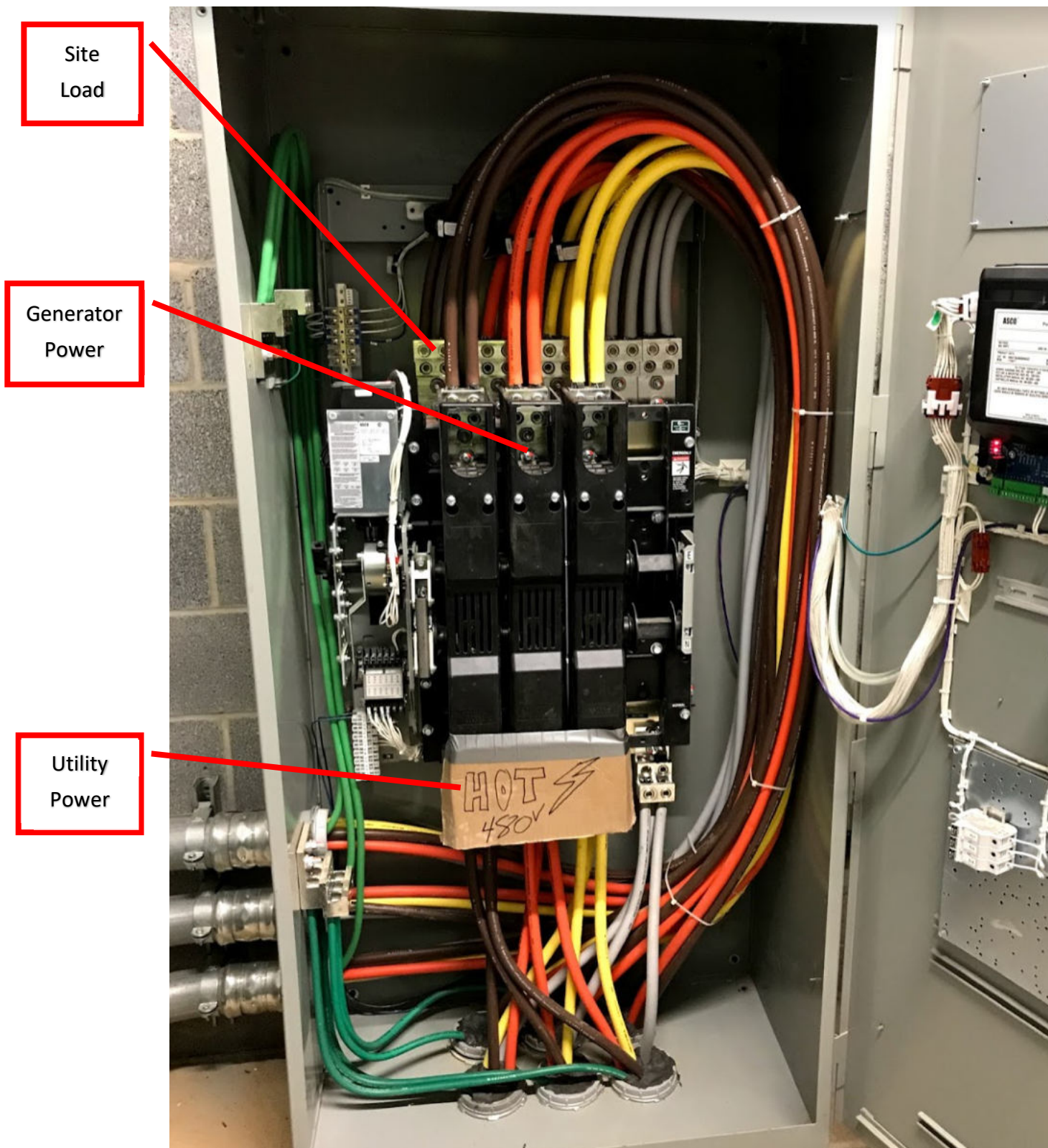
- ❑ The vent hood must be installed per these installation instructions.
- ❑ A room thermostat must control the unit. The use of fixed jumpers that will provide continuous heating or cooling is not allowed.
- ❑ A pre-filter must be installed at the entry to the return air duct.
- ❑ The return air duct must be provided and sealed to the unit.
- ❑ Return air temperature range between 55°F (13°C) and 80°F (27°C) must be maintained.
- ❑ Air filters must be replaced and pre-filter must be removed upon construction completion.
- ❑ The input rate and temperature rise must be set per the unit rating plate.
- ❑ The heat exchanger, components, duct system, air filters and evaporator coil must be thoroughly cleaned following final construction clean-up.
- ❑ The unit operating conditions (including airflow, cooling operation, ignition, input rate, temperature rise and venting) must be verified according to these installation instructions.



2. ELECTRICIAN

a. Asco Automatic Transfer Switch

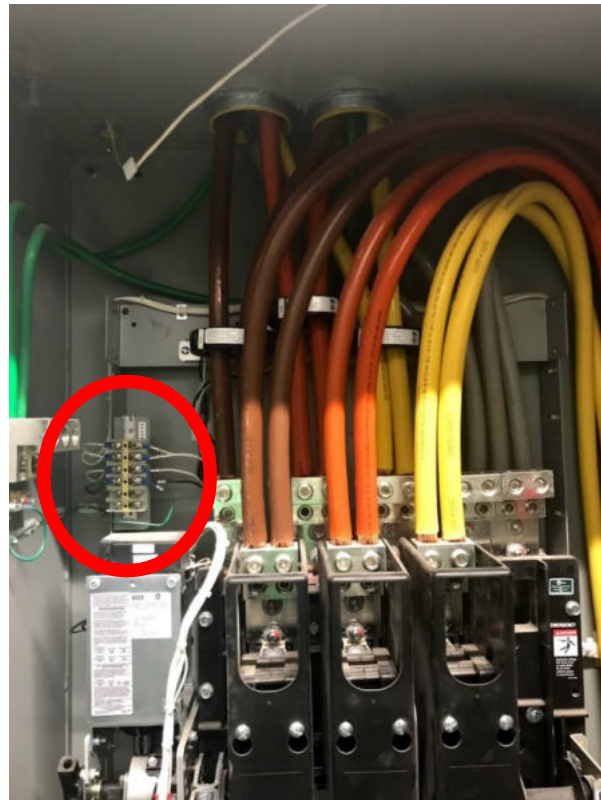
For proper operations the Asco Transfer Switch should be wired as shown below.



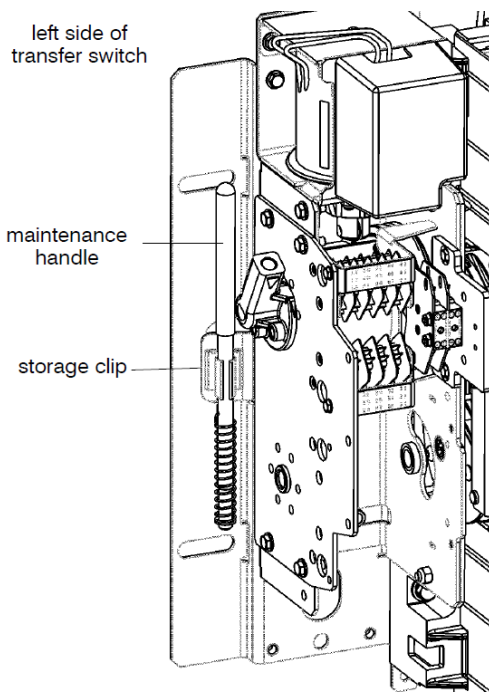


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Make sure the 6 brass safety grounding screws are removed from CT's wiring junction block.



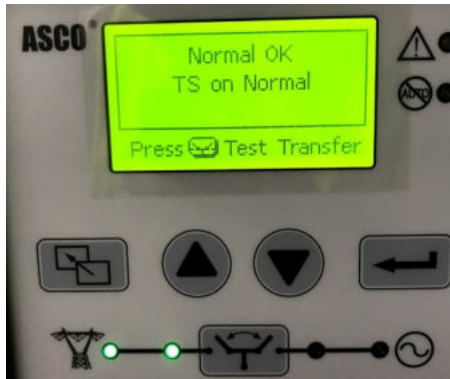
All ATS ship with Maintenance / Emergency Handle. Ensure handle is supplied and wiring is clear of manual switch throw.





Asco ATS (Auto Transfer Switch) Settings for Weekly Engine Exerciser.

Below is the main Asco ATS screen.



Press the double box button on the left to access Main Menu. Press the down arrow to access Settings. Press the right arrow button to enter Settings Menu.



In Settings press the down arrow to Engine Exerciser. And press the right arrow button to enter Engine Exerciser Menu.



In the Engine Exerciser Menu we will enable the proper settings below. You may be prompted for a password. It is 1111. The password screen will show 4 asterisk (****). Use the up/down arrow keys to change the number and right arrow to enter. Once the changes are accepted the selector box will have a check mark to confirm.





Settings:

Enable (Yes) ** Enables weekly testing**

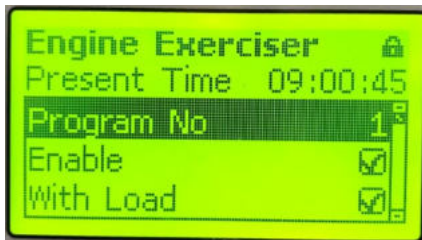
With Load (Yes) ** ATS will physically engage switch with building electrical load**

Start Week (All) ** Enables weekly testing for all weeks**

Day of Week (Mon) ** Enables weekly testing on Mondays only**

Time (06:30) ** Enables testing at 6:30 AM**

Duration (00:30) ** Enables test to run for 30 minutes only**



When complete press the left Double Box button to return to the Main Menu.





3. (MECHANICAL) HVAC RTU AND ACCESSORIES INSTALL

a. RTU 1 – MicroMetl Power Exhaust

Power Exhaust must be ordered from Lennox with Third Party (BMS) programmed controls for the Lenze VFD.

If not ordered properly you will need a secondary harness Part# 1002-LNZ-TPC.

Lennox dampers to be removed before Power Exhaust is installed.



Power Exhaust to have support legs added.





b. Lennox RTU 1, 3, 4, 5 & 6 (General Notes)

RTU 1 – Sales Middle – Economizer, Modulating power exhaust, Building Pressure Control, Dehumidification, CO2.

RTU 3 – Sales Front – Economizer, Standard power exhaust, CO2

RTU 4 - Bakery - No Economizer.

RTU 5 – Warehouse – Economizer, Standard power exhaust

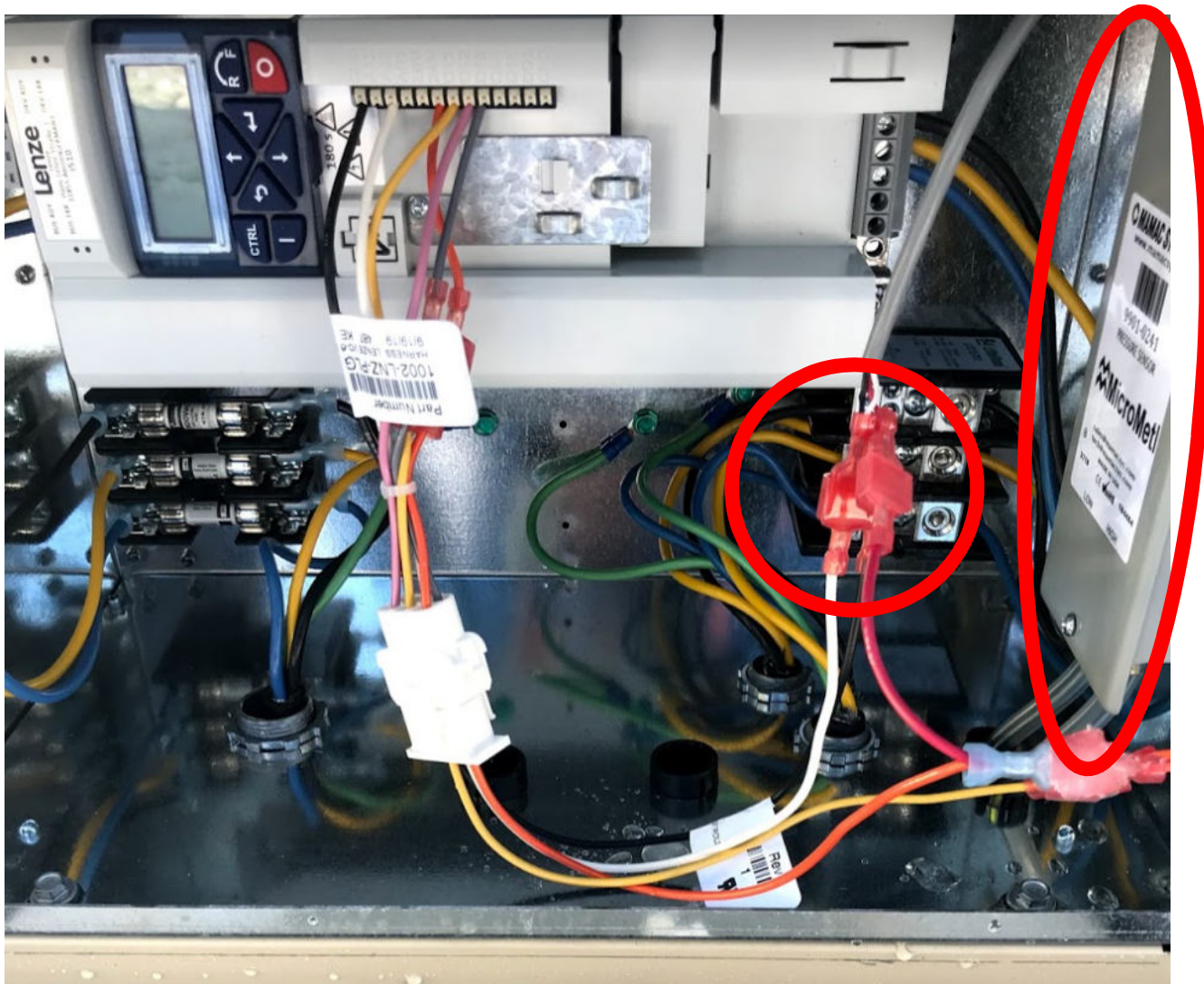
RTU 6 – Breakroom/Office – Economizer, Standard power exhaust

All units require inlet gas pressure regulator with pressure set to .35 psi or 10" w.c.

4. (CONTROLS/BMS)

a. RTU 1 – MicroMetl Power Exhaust and VFD

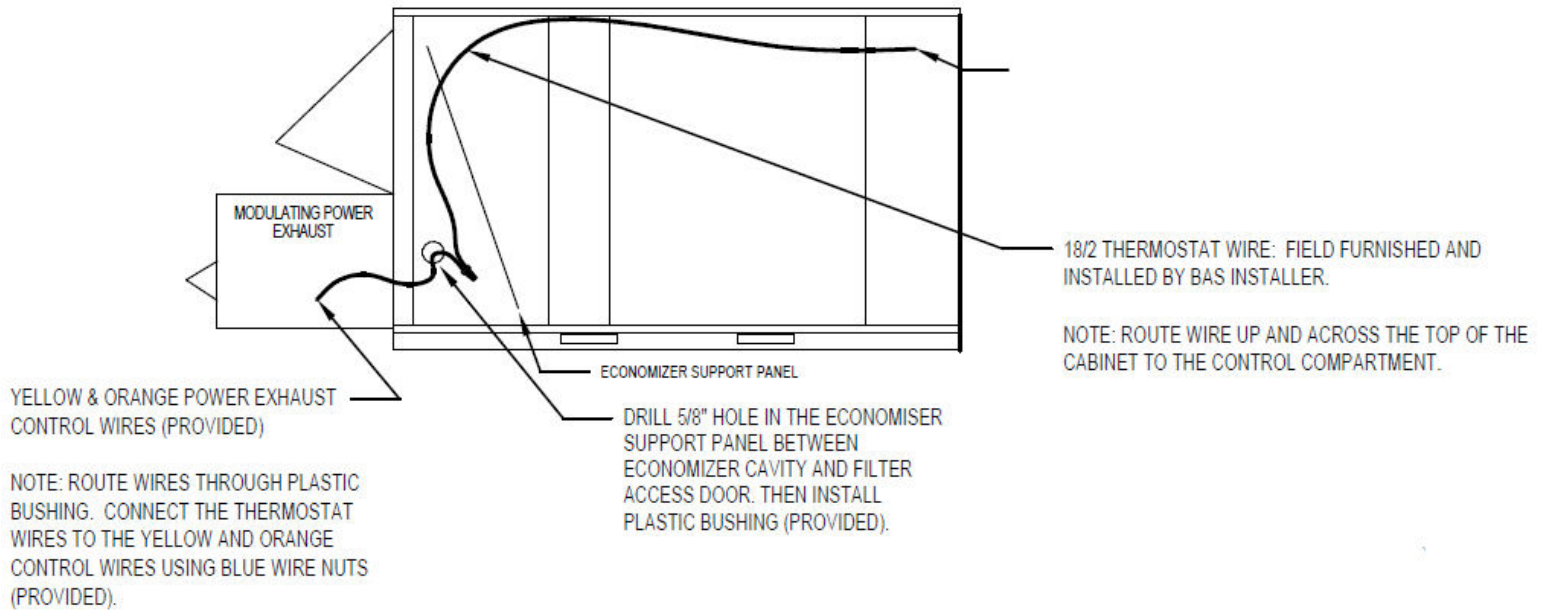
Pressure sensor switch located in Power Exhaust cabinet is not used. The 3 leads from the sensor to the VFD are to be disconnected and protected with electrical tape.





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Control point leads from the VFD are to be ran through the RTU 1 Lennox cabinet and landed on the Alerton VLC-853 control.



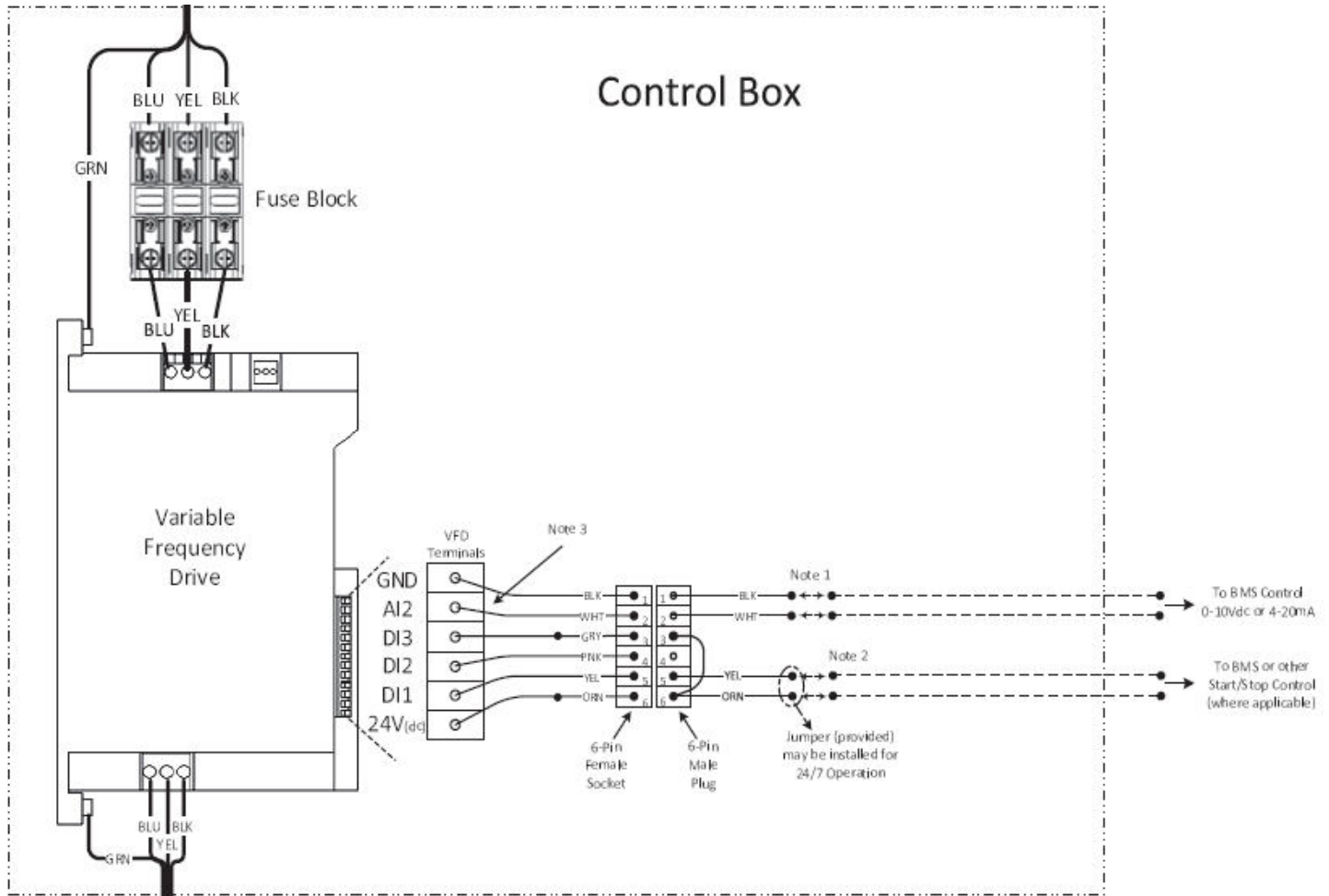


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Confirm the Power Exhaust was ordered from Lennox with Third Party (BMS) programmed controls for the Lenze VFD.

If not ordered properly you will need a secondary harness Part# 1002-LNZ-TPC.

**** Without the harness a jumpers can be placed across 24V to DI1 and 24V to DI3.****





b. RTU 1 – Accessories Power and Communication to BMS

120V to 24V / 40VA transformer to be installed in RTU 1 disconnect box.

Alerton, Setra and CO₂ sensor to be powered by 24V transformer

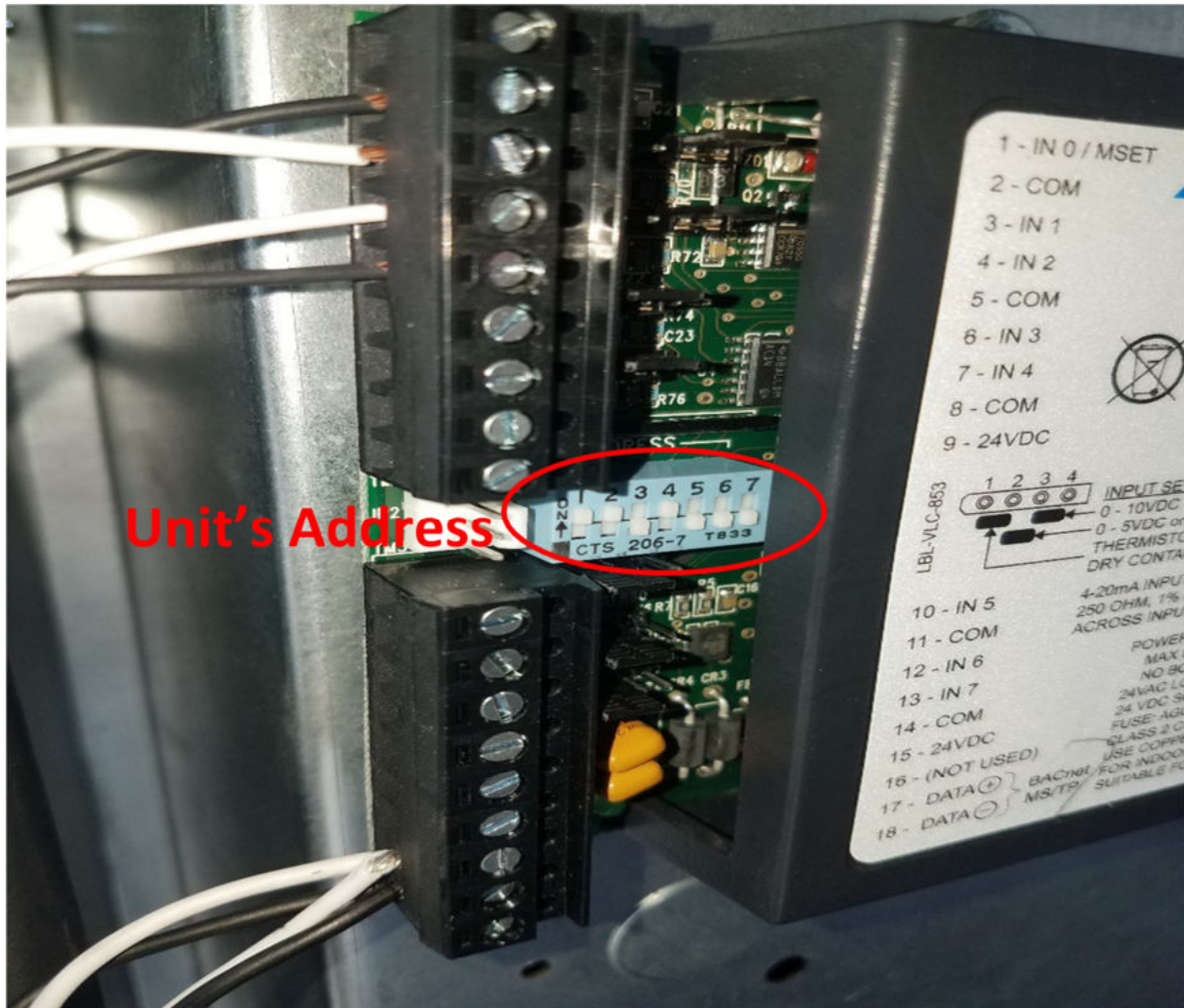
Transformer should be wired to 120V GFCI outlet circuit. (Photo below is for example only, as it shows a 480V to 24V stepdown transformer and is wired on RTU 1 disconnect and not GFCI outlet within RTU cabinet).





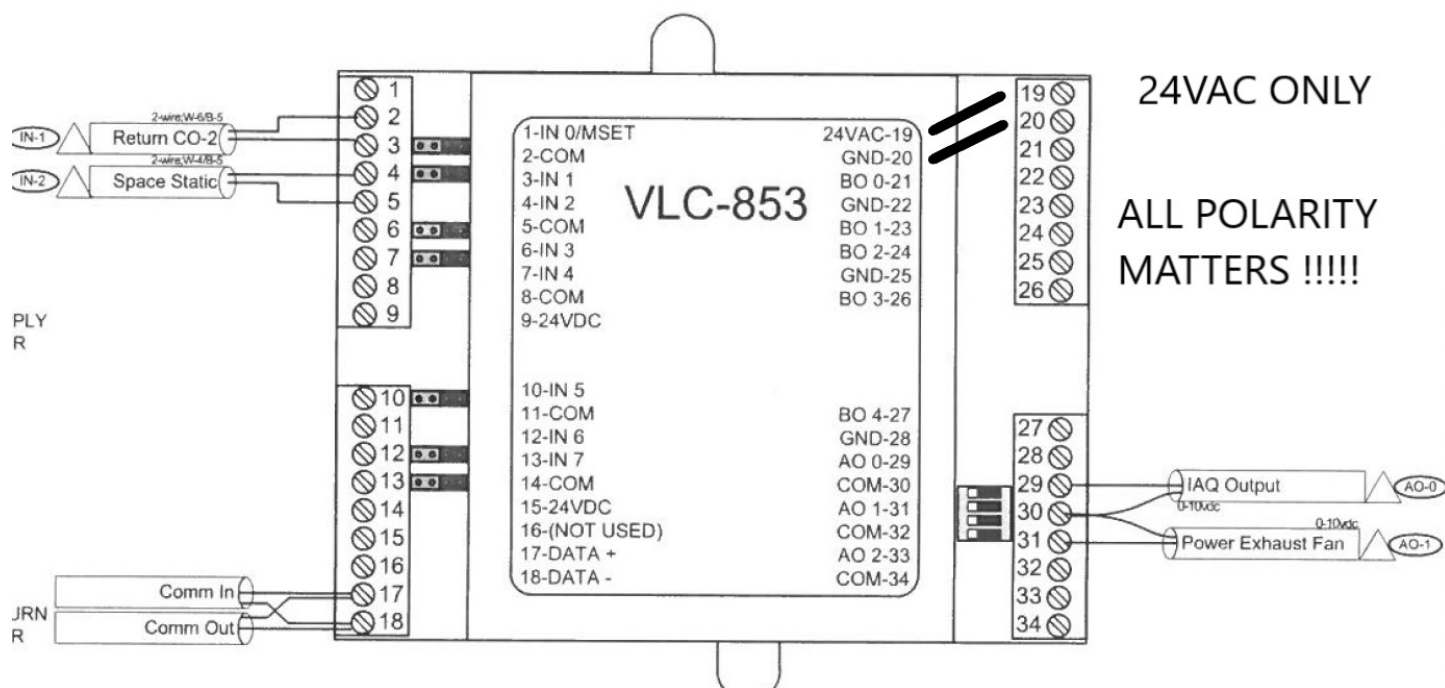
c. RTU 1 – Alerton VLC-853

Address the Alerton (Needs to be completed before initial Power Up)– Light blue switches on left. Switches (1), (2) and (4) in the UP position. All others DOWN.





Termination Wiring Points



- 24VAC power from Transformer
 - Terminal-19 (+)
 - Terminal-20 (-)
- Communication (BACnet)
 - Terminal-17: Common (-)
 - Terminal-18: Signal (+)
- Lennox CO-2
 - Terminal-2: Common (-)
 - Terminal-3: Signal (+)
- Setra Static PSI
 - Terminal-4: Signal (+)
 - Terminal-5: Common (-)
- IAQ Output to Lennox Prodigy Sensor IAQ Harness
 - Terminal-29: Signal (+)
 - Terminal-30: Common (-)
- Power Exhaust Fan
 - Terminal-31: Signal (+)
 - Terminal-30: Common (-)
- Jumpers for Input Signal at 0-10V
 - Terminal-3 jumper moved to right most 2 pin position
 - Terminal-4 jumper moved to right most 2 pin position





d. RTU 1 – Setra 2671MR2 Pressure Sensor

Verify that you have a model MR2 sensor.

Add a 4th ground wire from power/transformer to negative/comm. (Center connector of harness). This will be ground reference for output signal.

Highlighted below are the dip switch settings for the MR2.

1. Use the ± 0.25 dip switch settings for the 0-10v output. (Up, Down, Up, Down, Down)

NOTE: Black square in diagram indicates switch position.

MR1 range	MR2 range	MR3 range	MR4 range	Switch settings 0-5V output	Switch settings 0-10V output
Factory default setting					
0-0.1"	0-1"	0-5"	0-30"		
	0-.5"	0-2.5"	0-15"		
	0-.25"	0-1.25"	0-7.5"		
± 0.05 "	$\pm .5$ "	± 2.5 "	± 15 "		
	$\pm .25$ "	± 1.25 "	± 7.5 "		
	± 1.25 "	$\pm .625$ "	± 3.75 "		





e. RTU 2 – Munters Sensors and BACnet

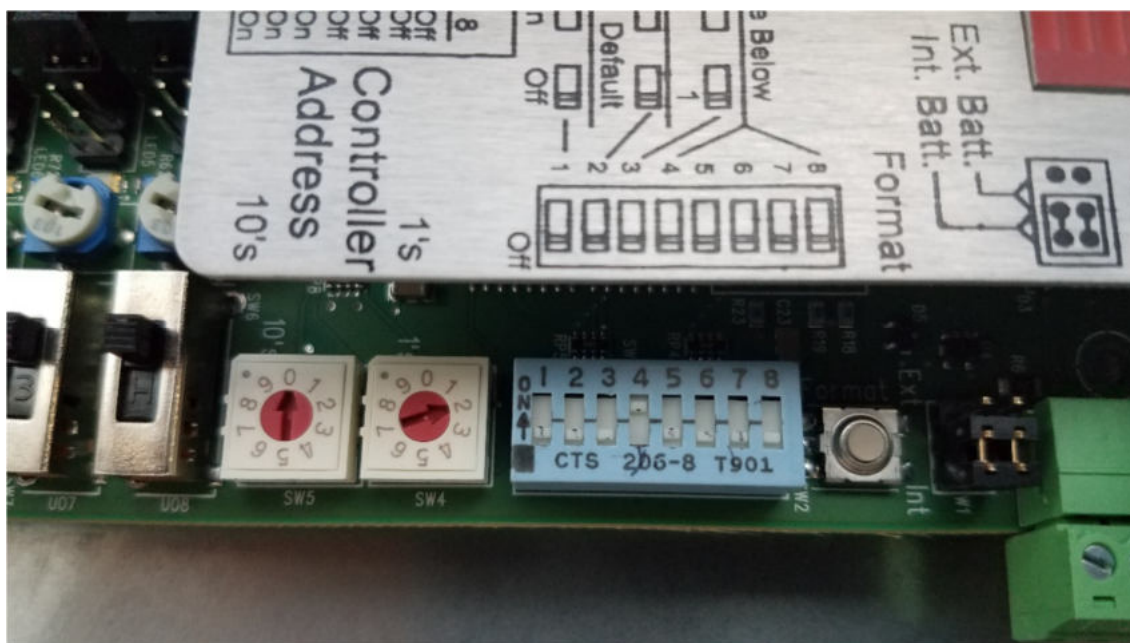
We have four sensors for the Munters unit.

1. Return air which is mounted in the unit. (Is factory mounted and should stay where it is at.)
2. The space combo which is the one with the white cover. (Is shipped loose and should be mounted on the exterior of the return duct.)
3. Supply air BAPI combo sensor. (Is shipped loose and should be mounted in the supply air duct work.)
4. And finally the BAPI button sensor. (Is shipped loose and should be mounted down on the sales floor pole/column.)

As for the wiring of the space combo and button sensor. The only part of the combo sensor that should be wired back to the unit is the humidity side. The button sensor is to be wired back to the unit where the temp side of the combo sensor would normally be tied in. The control program is set up to work on the input from the button sensor and the humidity side of the combo sensor that is mounted on the external of the return duct. The supply air combo sensor is used to control the heat modulation when the unit is in heating mode.

BACnet Port-1 settings and connections.

1. Rotary Dial settings SW5 (set to 0) SW4 (set to 2)
2. Dip Switch settings (4) UP all others Down.

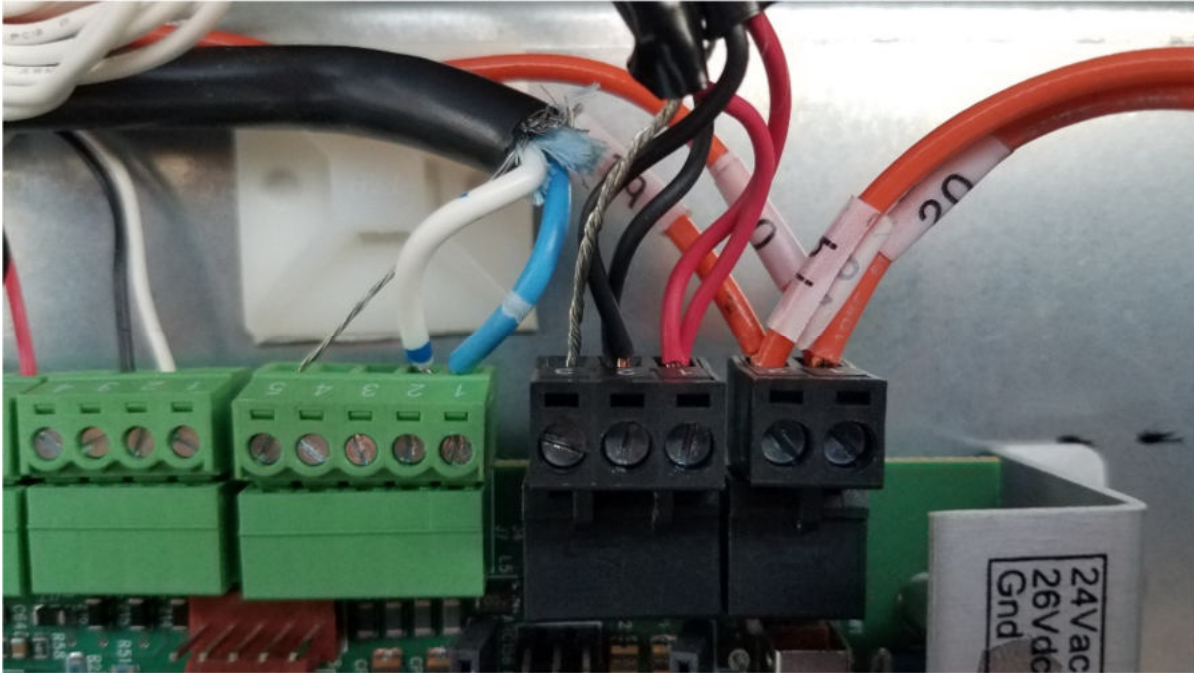




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The Munters BACnet network wiring for Port 1 is the Black (3) wire plug.

NOTE: You will need to move the Blue and White wires to the Green 5 wire plug to the left. Blue with white stripe (1). White with blue stripe (2). Shield to (5).

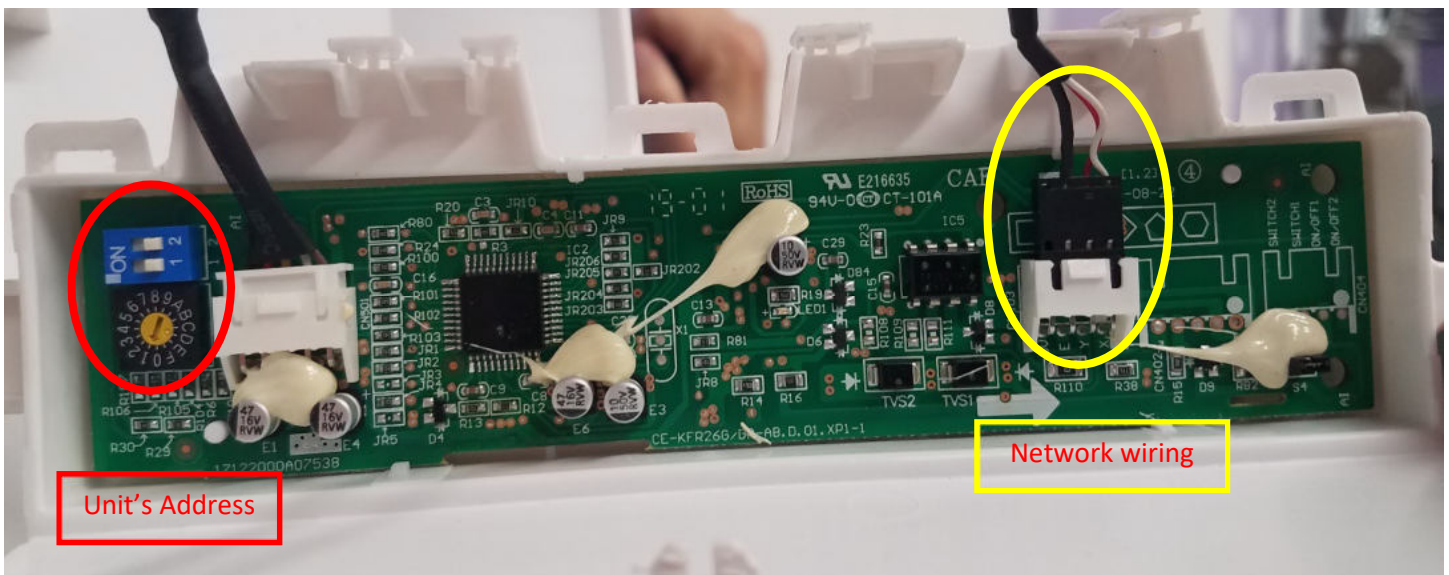
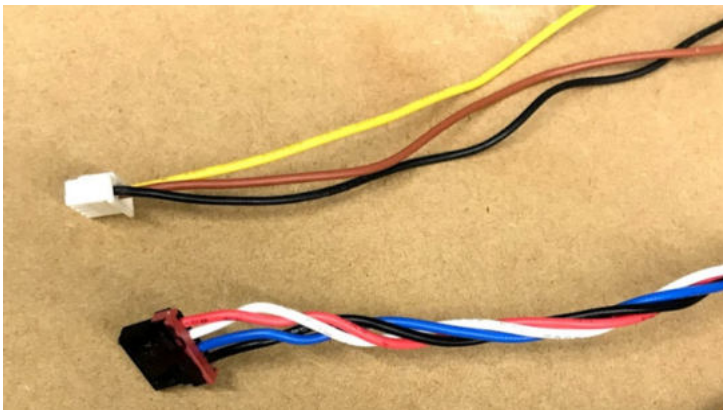




f. AC 1,2,3 – Split units BACnet wiring and addressing

NOTE:

1. **Mini-split wall-mounted indoor units are shipped with a cable for use with connecting to XYE terminal. Do not discard.**
2. (Pic 1) Make sure you have Network Harness Lennox Catalog Number 19P86 for network communication. (Black, Brown and Yellow harness below)
3. (Pic 2) If a separate network is pulled to each of the mini split units then **ALL** unit's rotary dial will be set to address (1).
4. If a single network is pulled and daisy chained to the unit's then the rotary dial will be addressed as follows: AC-1 (1), AC-2 (2) and AC-3 (3).





AC 1,2,3 – Split units BACnet wiring and Lennox Gateway

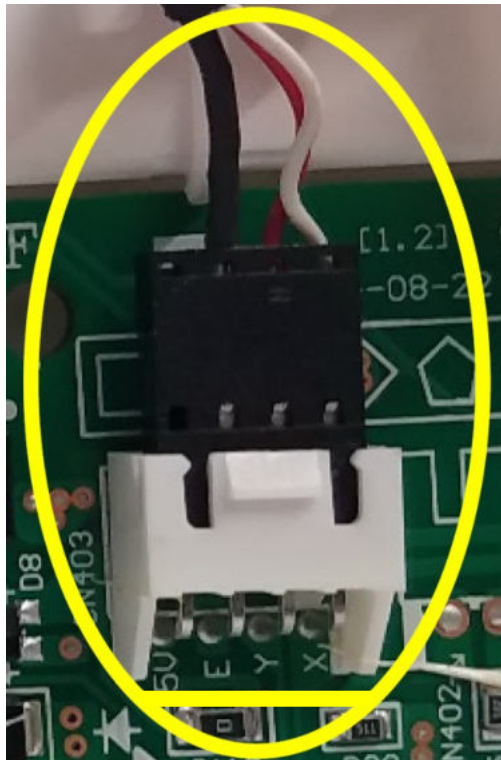
Network Harness Lennox Catalog Number 19P86 for network communication. (Black, Brown and Yellow harness).

Split Unit Wiring Points

X = F1

Y = F2

E = E



g. Onicon F5500 Gas Meter

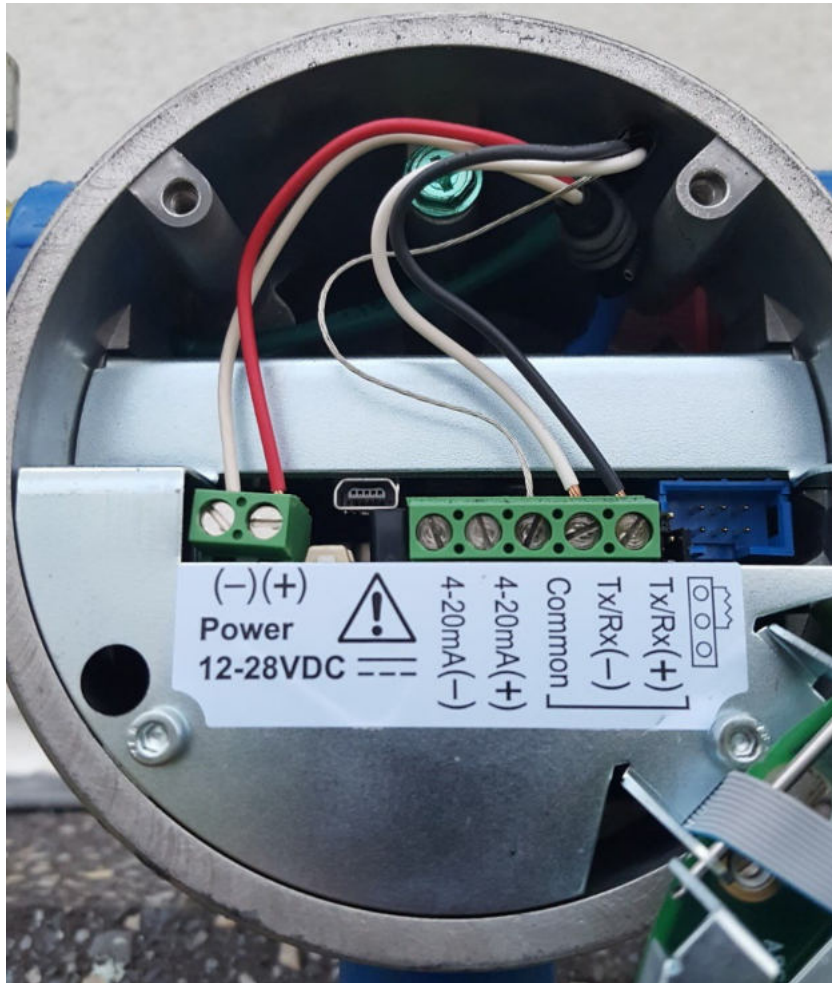
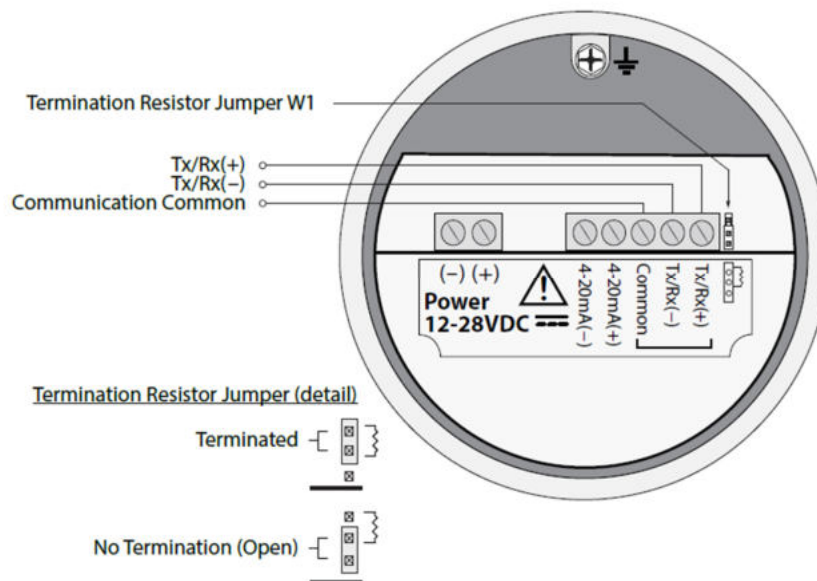


Fig. 3.6: RS485 Wiring





Settings to Use:

*DEFAULT LEVEL 1 PASSWORD = 1234

BAUD: 9600, PARITY: NONE, ADDRESS: (4)

Below are the steps to access the settings.

Modbus Communication Parameters

Communication Protocol and Parameters

To program the communication parameters, start at the Main Menu:

MAIN MENU			
I/O	FLO	DSP	EXIT
F1	F2	F3	F4

Then press **I/O (F1)** to set Inputs/Outputs:

SET I/O			
COM	420	EXIT	
F1	F2	F3	F4

Then press **COM (F1)** to select communication parameters.

Set Bus protocol for Modbus:

Comm=MODBUS			
NXT	OK		
F1	F2	F3	F4

Press **NXT (F1)** repeatedly until Modbus is selected as shown and then press **OK (F4)** to accept the setting.

The following communication parameters are only available for MODBUS:

Baud=9600			
NXT	OK		
F1	F2	F3	F4

Press **NXT (F1)** repeatedly until the correct selection is shown and then press **OK (F4)** to accept the setting.





Modbus Communication Parameters

Selections are:

- "115200"
- "57600"
- "38400"
- "19200"
- "9600"
- "4800"
- "2400"
- "1200"

Parity=EVEN
NXT OK

F1 F2 F3 F4

Press **NXT (F1)** repeatedly until the correct selection is shown and then press **OK (F4)** to accept the setting.

Selections are: "NONE"
"ODD"
"EVEN"

Address=02
CHG OK

F1 F2 F3 F4

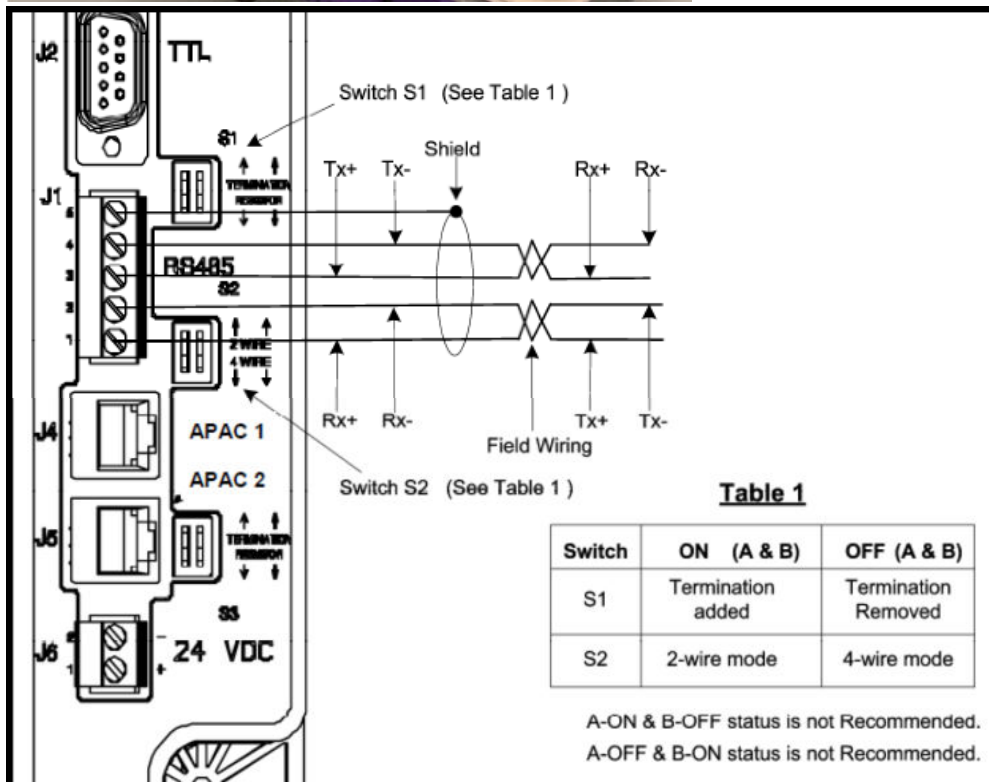
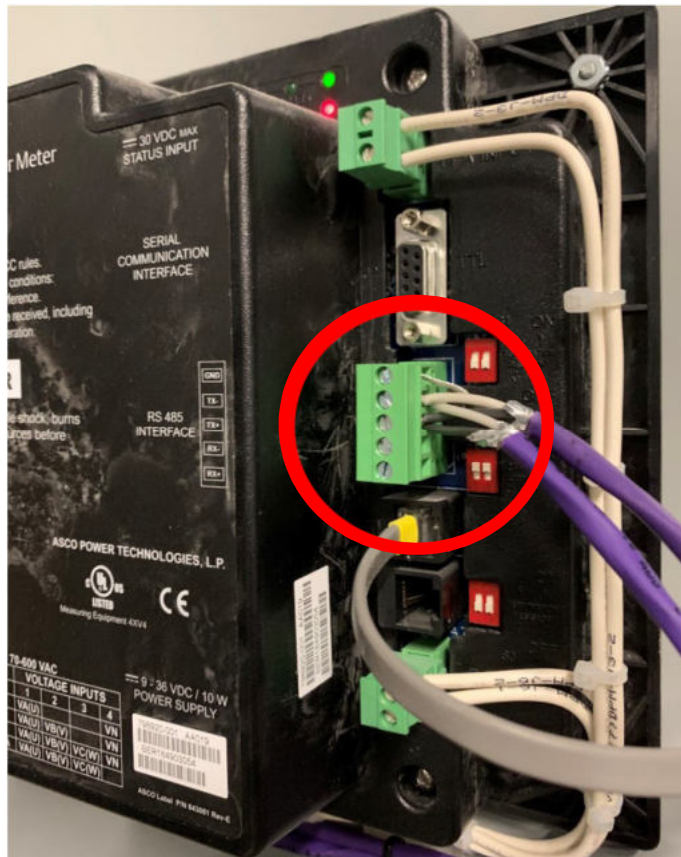
Press **CHG (F1)** to change the address and then press **OK (F4)** to accept the setting.

Selections are between 01 and 247.

Note: Power cycle is required for the new settings to take effect.



h. Asco Energy Meter



SETTINGS TO USE: 485 SETUP: 485 PROTOCOL: MODBUS, 485 ADDRESS: (3), 485 BAUD: 9600

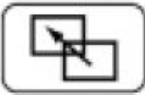
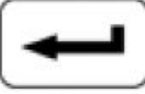
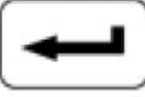
RS485 Serial Communication Interface port J1

If the Digital Power Meter is connected to a communications network via the RS485 (J1) port, select the appropriate protocol, baud rate, and address for the port as follows:

Protocol

- **ASCO II** – Enters the Digital Power Meter in an ASCO I/O Module emulation mode when used on ASCO ATSS and *PowerQuest* devices.
- **Modbus RTU**– Choose this selection when the Digital Power Meter is to be used on a network that communicates via the Modbus RTU protocol. Contact ASCO Power Technologies to obtain a document detailing the corresponding Modbus protocol Register map definitions.

Baud Rate	9600, 19200, 38400, 57600 bps	Address	1-239 (unique for each Digital Power Meter)
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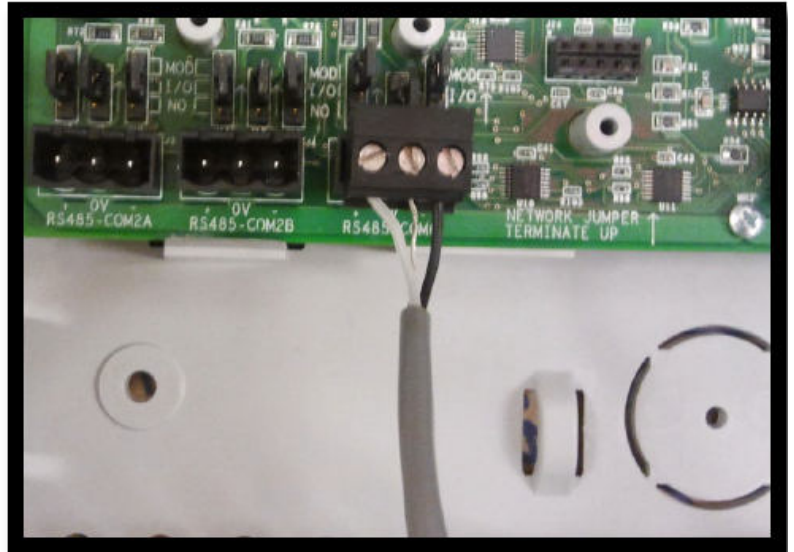
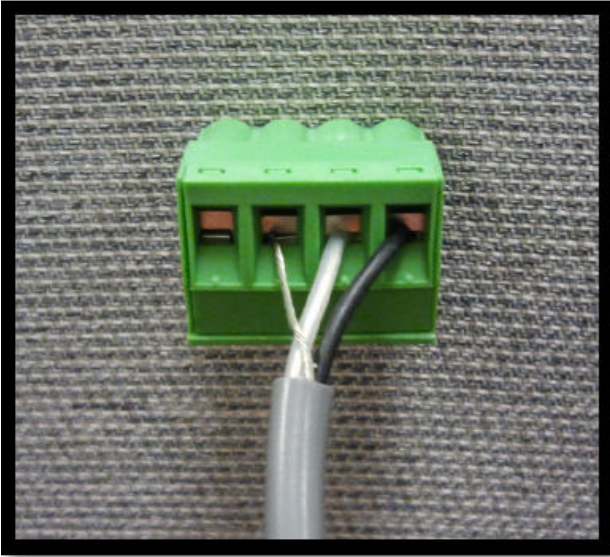
Step	Press	Display Shows	Comment
1		240V 3.601kW 0.000kW 3.600kVA 5A +1.0PF #0.0Hz 0.0Hrs	Displays the overview screen if not already shown.
2		SETTINGS General Communication Display Service	The Settings screen is displayed. Press the up and down arrow keys until the Communication option is highlighted.
3		COMMUNICATION 485 Protocol: ASCOII 485 Address: 1 485 Baudrate: 19200	Press the up and down arrow keys until the 485 Protocol option is highlighted.
4		COMMUNICATION 485 Protocol: ASCOII 485 Address: 1 485 Baudrate: 19200	485 Protocol is blinking.
5		COMMUNICATION 485 Protocol: MODBUS 485 Address: 1 485 Baudrate: 19200	Press the up and down arrow keys and select the desired protocol.
6		COMMUNICATION 485 Protocol: MODBUS 485 Address: 1 485 Baudrate: 19200	Press the enter key to save the setting.
7		COMMUNICATION 485 Protocol: MODBUS 485 Address: 1 485 Baudrate: 19200	Repeat steps 5 and 6 for the 485 Address and 485 Baud Rate.

Now press the  escape key to return to the top level display.

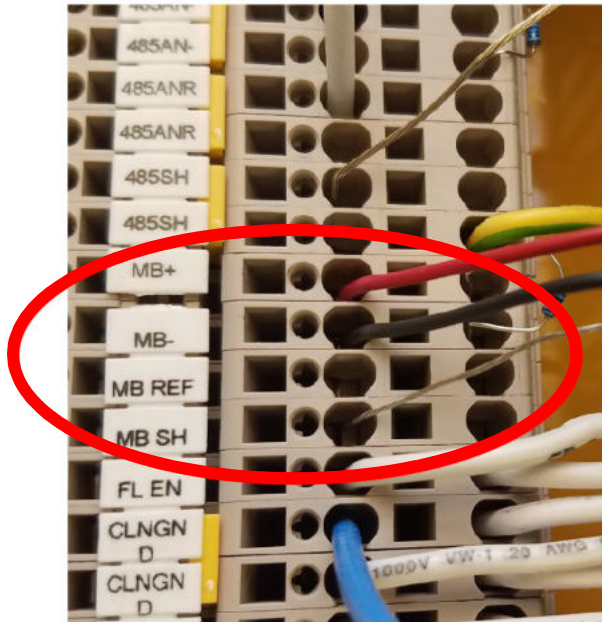


i. Caterpillar Generator

Network wiring if terminating on CAT control module: Data A to -, Data B to +, Shield to gnd. (Reversed at E2)

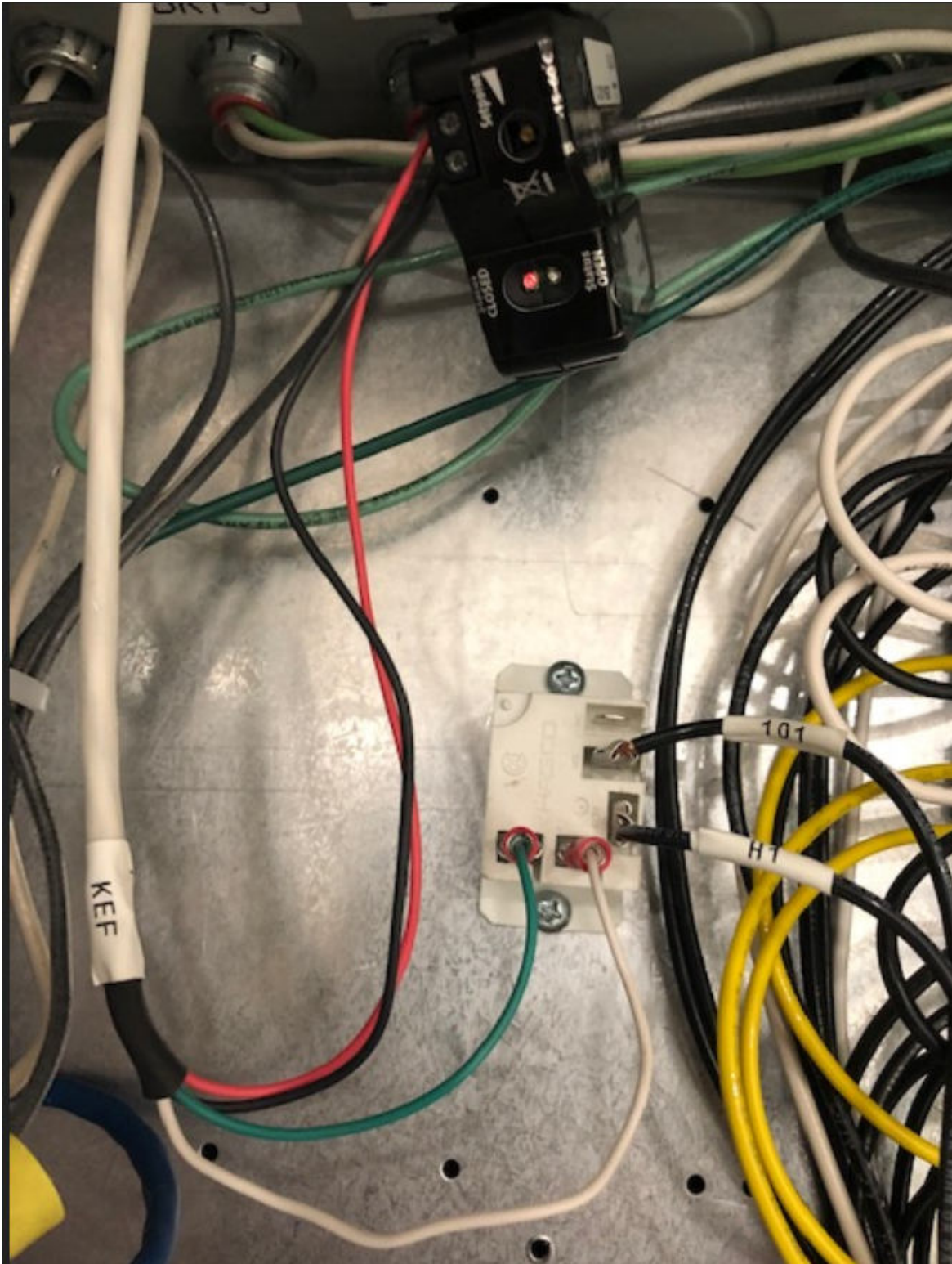


Network Wiring if terminating on terminal strip: MB+, MB-, MB SH. (Reversed at E2)

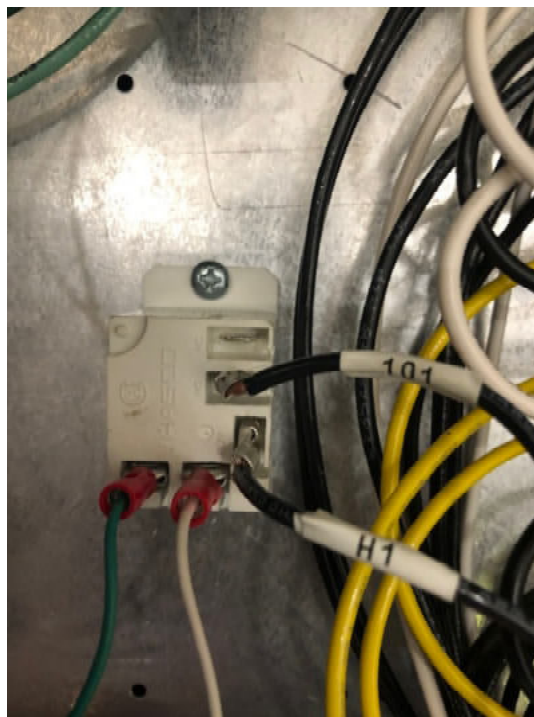


j. CaptiveAire Kitchen Exhaust Fan

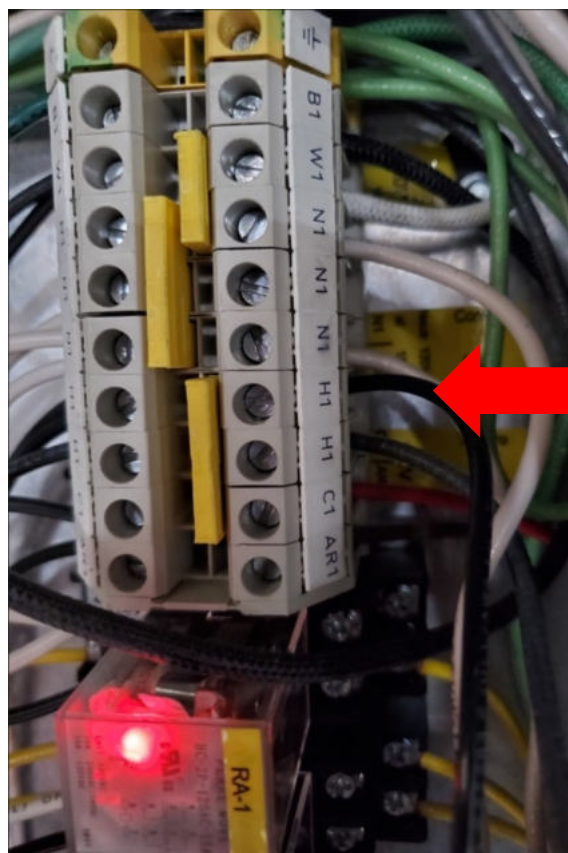
Below are steps and photos for CT and relay install location in hood. And wiring points to relay for proof.



24V wire (green / white) from Emerson E2 output (N / O) to the relay coil.



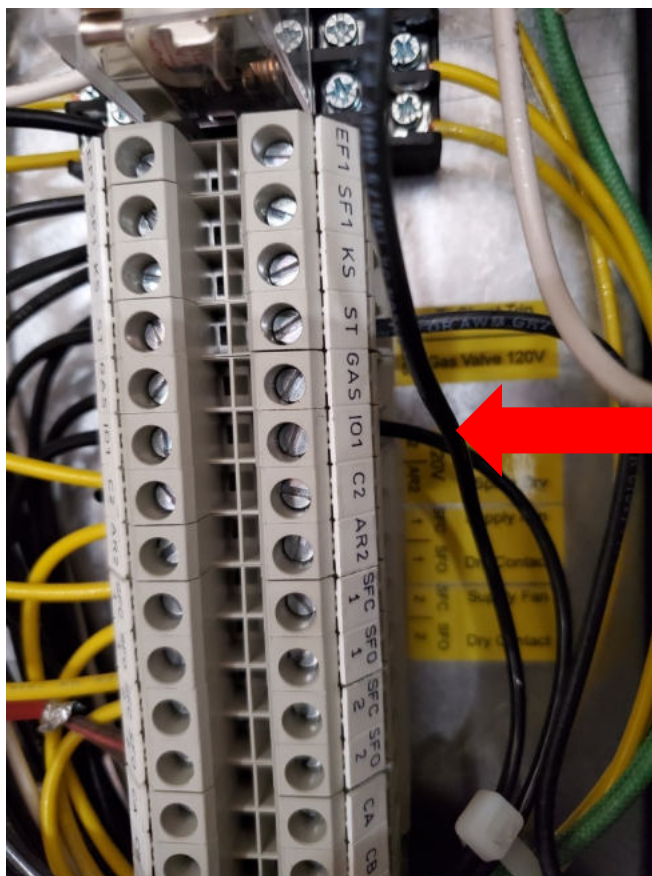
Install wire from the H1 termination block within the hood to the common on the relay.





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Install wire from I01 termination block within the hood to the (N/O) on the relay.





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k. Transformer add to lighting contactor for Emerson control.

120V to 24V 50VA transformer to be used to power up Emerson relay terminal blocks with 24VAC.

Same power can be used to feed relay out for KEF-1 and TEF's 1,2,3 enable relays.

